

COMPONENTI ELETTRICI

mercoledì 2 marzo 2022 18:04

→ PASSIVI : $\bar{E}_q = \int_0^t p(\tau) d\tau \geq 0 \quad \forall t$

→ ATTIVO ; NON RISPETTA ↗

→ CONTROLLATO IN TENSIONE : $i = f(v)$

→ CONTROLLATO IN CORRENTE : $v = f(i)$

→ CON MEMORIA : LA TENSIONE E LA CORRENTE ALL'ISTANTE t DIPENDONO DA GRANDEZZE AGLI ISTANTI PRECEDENTI

→ SENZA MEMORIA : LA TENSIONE E LA CORRENTE ALL'ISTANTE t DIPENDONO DA GRANDEZZE ALL'ISTANTE t

RESISTORE

mercoledì 2 marzo 2022 18:15



EQ. COSTITUTIVA: (1^a LEGGE DI OHM)

$$v = R i$$

$\rightarrow$ RESISTENZA $[\Omega]$

$$R = \rho \cdot \frac{l}{S}$$

ρ RESISTIVITA' DEL MATERIALE $[\Omega \cdot m]$ (2^a LEGGE DI OHM)
 l $\rightarrow$ LUNGHEZZA
 S $\rightarrow$ SEZIONE

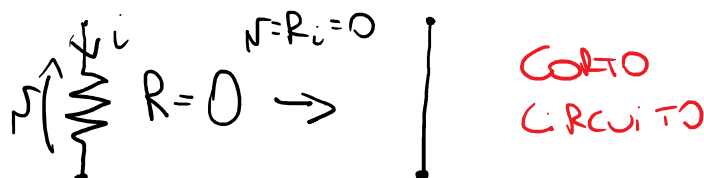
- POTENZA: $p = v \cdot i = R i^2 = \frac{v^2}{R}$ LEGGE DI JOULE

p SEMPRE $\geq 0 \Rightarrow$ COMPONENTE PASSIVO

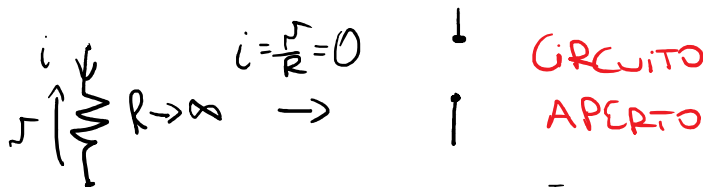
$v = R i \rightarrow$ CONTROLLATO IN CORRENTE

$i = \frac{v}{R} \rightarrow$ CONTROLLATO IN TENSIONE

$v(t) = R i(t) \rightarrow$ SENZA MEMORIA



CORTO
CIRCUITO

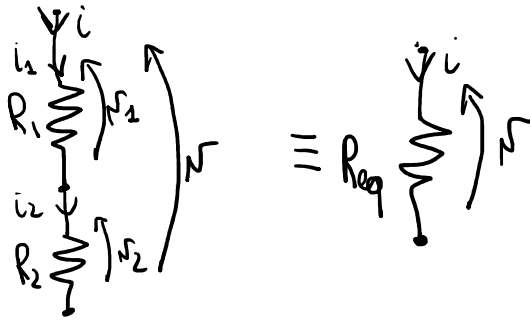


CIRCUITO
APERTO

ENERGIA: $w(t_1, t_2) = \int_{t_1}^{t_2} p(\tau) d\tau$

COLLEGAMENTO IN SERIE

mercoledì 2 marzo 2022 18:30



$$\begin{aligned} V &= V_1 + V_2 = R_1 i_1 + R_2 i_2 = \\ &= R_1 i + R_2 i = \underbrace{(R_1 + R_2)}_{R_{eq}} i \end{aligned}$$

DUE O PIÙ RESISTORI SONO IN SERIE QUANDO SONO
ATTRAVERSATI DALLA STESSA CORRENTE

$$R_{eq} = \sum_{i=1}^{\text{NUMERO DI RES. IN SERIE}} R_i$$

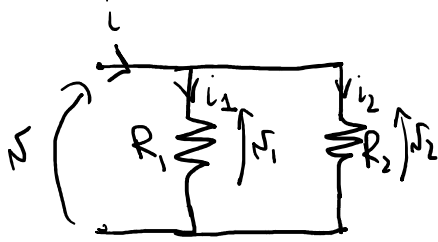
$$V = R_{eq} i \Rightarrow i = \frac{V}{R_{eq}} = \frac{V}{R_1 + R_2}$$

$$\begin{aligned} \rightarrow V_1 &= R_1 i = \frac{R_1}{R_1 + R_2} V \\ \rightarrow V_2 &= R_2 i = \frac{R_2}{R_1 + R_2} V \end{aligned}$$

PARTITORE
Di
TENSIONI

COLLEGAMENTO IN PARALLELO

mercoledì 2 marzo 2022 18:39



DUE O PIÙ RESISTORI SI DICONO IN PARALLELO QUANDO AI LORO CAPI VI È LA STESSA TENSIONE



$$I_1 = \frac{V_1}{R_1} \quad I_2 = \frac{V_2}{R_2}$$

$$I = I_1 + I_2 = \frac{V_1}{R_1} + \frac{V_2}{R_2} = \frac{V}{R_1} + \frac{V}{R_2} = \underbrace{\left(\frac{1}{R_1} + \frac{1}{R_2} \right)}_{1/R_{eq}} V$$

$$R_{eq} = \left(\sum_{i=1}^n \frac{1}{R_i} \right)^{-1}$$

$$\frac{1}{R_{eq}} = \frac{R_1 + R_2}{R_1 R_2} \Rightarrow R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

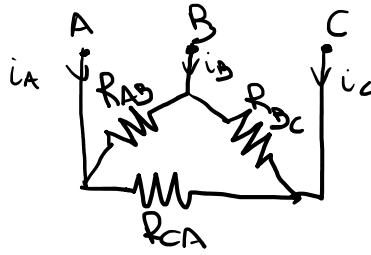
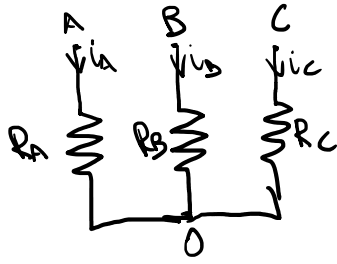
$$V = R_{eq} \cdot I = \frac{R_1 R_2}{R_1 + R_2} I$$

$$\begin{aligned} I_1 &= \frac{V_1}{R_1} = \frac{V}{R_1} = \frac{R_2}{R_1 + R_2} I \\ I_2 &= \frac{V_2}{R_2} = \frac{V}{R_2} = \frac{R_1}{R_1 + R_2} I \end{aligned}$$

PARTITORI
DI
CORRENTI

COLLEGAMENTO STELLA-TRIANGOLO

venerdì 4 marzo 2022 09:21



Supponiamo di scollegare il terminale C:

$$\textcircled{1} R_A + R_B = R_{AB} \parallel (R_{BC} + R_{CA})$$

scollegiamo B:

$$\textcircled{2} R_A + R_C = R_{CA} \parallel (R_{AB} + R_{BC})$$

scollegiamo A:

$$\textcircled{3} R_B + R_C = R_{BC} \parallel (R_{AB} + R_{CA})$$

$$R_A = \frac{R_{AB} \cdot R_{CA}}{R_{AB} + R_{BC} + R_{CA}}$$

$$R_B = \frac{R_{BC} \cdot R_{AB}}{R_{AB} + R_{BC} + R_{CA}}$$

$$R_C = \frac{R_{CA} \cdot R_{BC}}{R_{AB} + R_{BC} + R_{CA}}$$

$$R_{AB} = R_{BC} = R_{CA} = R_{\Delta} \Rightarrow R_Y = \frac{R_{\Delta}}{3}$$

$$R_{AB} = \frac{R_A R_B + R_B R_C + R_C R_A}{R_C}$$

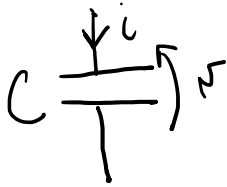
$$R_{BC} = \frac{R_A R_B + R_B R_C + R_C R_A}{R_A}$$

$$R_{CA} = \frac{R_A R_B + R_B R_C + R_C R_A}{R_B}$$

$$R_A = R_B = R_C = R_Y \Rightarrow R_{\Delta} = 3 R_Y$$

CONDENSATORE

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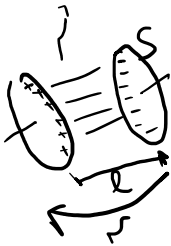


EQ. COSTITUTIVA:

$$i = C \frac{dv}{dt}$$

CAPACITÀ [F]

DIELETTRICO



$$Q = C V$$

$$i = \frac{dQ}{dt} = C \frac{dv}{dt}$$

PERMITTIVITÀ DEL MATERIALE

$$C = \epsilon \cdot \frac{S}{d}$$

S → SEZIONE
d → DISTANZA

- $p(t) = v(t) \cdot i(t) = v(t) \cdot C \frac{dv}{dt}$

POT. ASSORBITA
POT. EROGATA

- $w(t_2, t_1) = \int_{t_1}^{t_2} p(t) dt = \int_{t_1}^{t_2} v(t) \cdot C \frac{dv}{dt} dt = C \int_{v_1}^{v_2} v(t) dv = \frac{1}{2} C [v^2(t_2) - v^2(t_1)]$

ENERGIA
NON DISSIPATIVO

$t_1 = 0 \rightarrow w = \frac{1}{2} C [v^2(t_2) - v^2(0)]$

PASSIVO

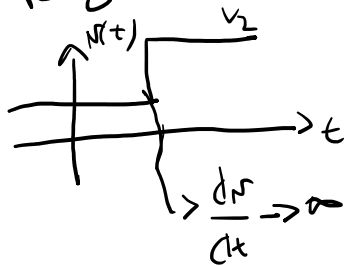
LA TENSIONE DEFINISCE LO STATO ENERGETICO $\Rightarrow$ VARIABILE DI STATO

- $v(t) = \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau = \underbrace{\frac{1}{C} \int_{-\infty}^0 i(\tau) d\tau}_{v(0)} + \frac{1}{C} \int_0^t i(\tau) d\tau$

COND. INIZIALI
MEMORIA

$$i(t) = C \frac{dv}{dt} \rightarrow \int \frac{dv}{dt} = \frac{1}{C} i(t)$$

Supponiamo:



$$i = C \frac{dv}{dt} \rightarrow \infty \quad p = v(t) \cdot i(t) \rightarrow \infty \quad \text{NON È POSSIBILE}$$

$\Rightarrow$ LA TENSIONE NON PUÒ VARIARE BRUSCAMENTE

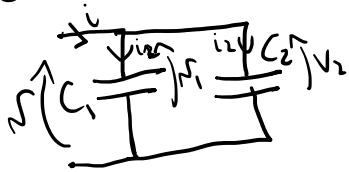
$$v(t) = V \rightarrow i(t) = C \frac{dv}{dt} = 0$$

↓
CIRCUITO
APERTO

Condensatore cilindrico:



Collegamento in parallelo:



$$C_{eq} = \sum_{i=1}^N C_i$$

$$i = i_1 + i_2 = C_1 \frac{dv}{dt} + C_2 \frac{dv}{dt} = \overbrace{(C_1 + C_2)}^{C_{eq}} \frac{dv}{dt}$$

Collegamento in serie

$$C_1 \xrightarrow{i_1} C_2 \xrightarrow{i_2} \equiv \left(\frac{1}{C_1} + \frac{1}{C_2} \right)^{-1} C_{eq}$$

$$i = C_{eq} \frac{dv}{dt} = C_{eq} \left(\frac{dv_1}{dt} + \frac{dv_2}{dt} \right) = C_{eq} \left(\frac{i_1}{C_1} + \frac{i_2}{C_2} \right) = C_{eq} \left(\frac{1}{C_1} + \frac{1}{C_2} \right) i$$

$$C_q \left(\frac{1}{C_1} + \frac{1}{C_2} \right) = 1$$

$$\Rightarrow C_q = \frac{1}{\left(\frac{1}{C_1} + \frac{1}{C_2} \right)} = \left(\frac{1}{C_1} + \frac{1}{C_2} \right)^{-1}$$

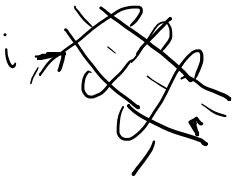
Induttore

venerdì 4 marzo 2022 10:25



EQ. CARATTERISTICA
 $\rightarrow$ INDUTTANZA [H]

$$v = L \frac{di}{dt}$$



$$\phi = L i$$

$$N \frac{d\phi}{dt} = L \frac{di(t)}{dt}$$

$$L = \frac{\mu \cdot S \cdot N^2}{l}$$

$\rightarrow$ PERMEABILITÀ DEL MATERIALE
 $\rightarrow$ NUMERO DI SPIRE
 $\rightarrow$ SEZIONE DELLA SPIRA
 $\rightarrow$ LUNGHEZZA

$$i(t) = \frac{1}{L} \int_{-\infty}^t v(\tau) d\tau = \underbrace{\frac{1}{L} \int_{-\infty}^0 v(\tau) d\tau}_{i(0)} + \frac{1}{L} \int_0^t v(\tau) d\tau = i(0) + \frac{1}{L} \int_0^t v(\tau) d\tau$$

CON MEMORIA

$$p(t) = v(t) \cdot i(t) = L \frac{di(t)}{dt} \cdot i(t)$$

$\begin{cases} P > 0 & \rightarrow \text{POT. ASSORBITA} \\ P < 0 & \rightarrow \text{POT. EROGATA} \end{cases}$

NON DISSIPATIVO

ENERGIA

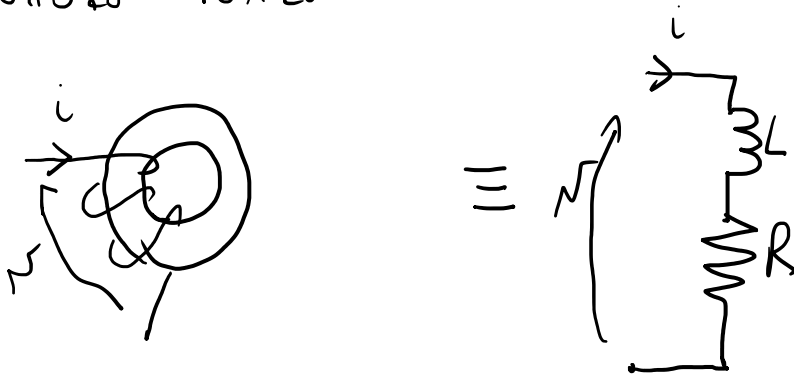
$$w(t_1, t_2) = \int_{t_1}^{t_2} p(t) dt = \int_{t_1}^{t_2} L i(t) \frac{di}{dt} dt = L \int_{i_1}^{i_2} i(t) di = \frac{1}{2} L \left[I^2 \right]_{i_1}^{i_2} = \frac{1}{2} L [I^2(t_2) - I^2(t_1)]$$

$$w(0, t_2) = \frac{1}{2} L [I^2(t_2) - I^2(0)]$$

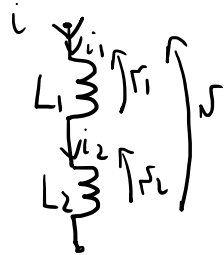
PASSIVO

LA CORRENTE INDICA LO STATO ENERGETICO $\rightarrow$ VARIABILE DI STATO

INDUTTORI REALI



COLLEGAMENTO SERIE



$$L_{eq} = \sum_{i=1}^n L_i$$

$$\mathcal{N} = \mathcal{N}_1 + \mathcal{N}_2 = L_1 \frac{di_1}{dt} + L_2 \frac{di_2}{dt} = \overbrace{(L_1 + L_2)}^{L_{eq}} \frac{di}{dt}$$

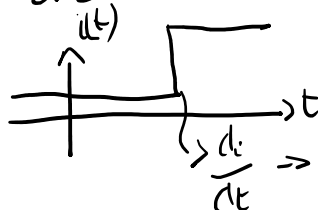
COLLEGAMENTO PARALLELO



$$L_{eq} = \left(\sum_{i=1}^n \frac{1}{L_i} \right)^{-1}$$

$$\begin{aligned} \mathcal{N} &= L_{eq} \cdot \frac{di}{dt} = L_{eq} \left(\overset{\sim \frac{\mathcal{N}_1}{L_1}}{\frac{di_1}{dt}} + \overset{\sim \frac{\mathcal{N}_2}{L_2}}{\frac{di_2}{dt}} \right) = L_{eq} \left(\frac{\mathcal{N}_1}{L_1} + \frac{\mathcal{N}_2}{L_2} \right) = \underbrace{L_{eq} \left(\frac{1}{L_1} + \frac{1}{L_2} \right)}_{=1} \mathcal{N} \\ \Rightarrow L_{eq} &= \left(\frac{1}{L_1} + \frac{1}{L_2} \right)^{-1} \end{aligned}$$

Supponiamo:



$$\frac{di}{dt} \rightarrow \infty \Rightarrow \mathcal{N} = L \frac{di}{dt} \rightarrow \infty \Rightarrow p(t) = \mathcal{N}(t) \cdot i(t) \rightarrow \infty$$

Non
è
possibile

$$\text{---} \quad \left(\frac{di}{dt} \rightarrow \infty \Rightarrow v = L \frac{di}{dt} \rightarrow \infty \Rightarrow p(t) = v(t) \cdot i(t) \rightarrow \infty \right. \quad \begin{array}{l} \text{É} \\ \text{POSSIBILE} \end{array}$$

NON POSSIAMO AVERE VARIAZIONI ISTANTANEE DELLA
CORRENTE

$$i(t) = I \rightarrow v = L \frac{di}{dt} = 0 \Rightarrow \left[\begin{array}{l} \text{CORSO IN} \\ \text{CORTO-CIRCUITO} \end{array} \right.$$

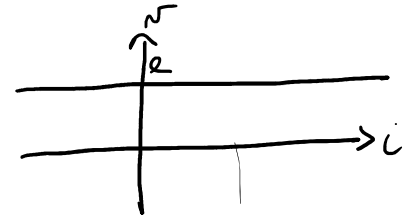
GENERATORE INDIPENDENTE DI TENSIONE

mercoledì 9 marzo 2022 16:17



EQ. COSTITUTIVA

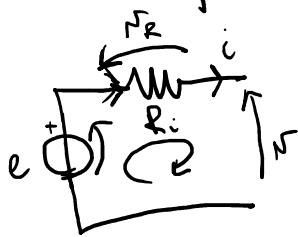
$$v = e \quad \forall t, i$$



POTENZA: $p(t) = v(t) \cdot i(t) = e \cdot i$

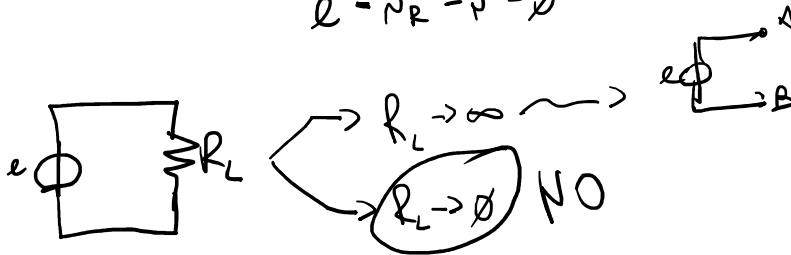
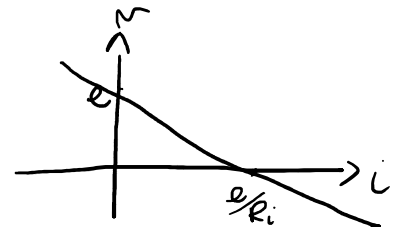
SIA POSITIVA CHE NEGATIVA

$Q = 0 \rightarrow$ CORTO CIRCUITO



$$v = e - R_i \cdot i$$

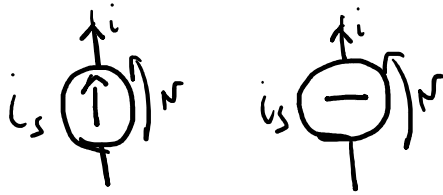
$$e - v_{R_L} - v = 0$$



$R_L \rightarrow \infty$
 $R_L \rightarrow 0$ NO

GENERATORE INDIPENDENTE DI CORRENTE

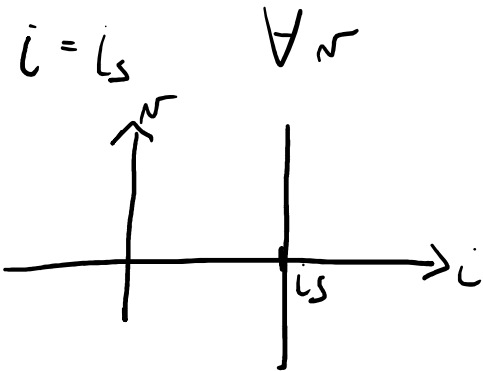
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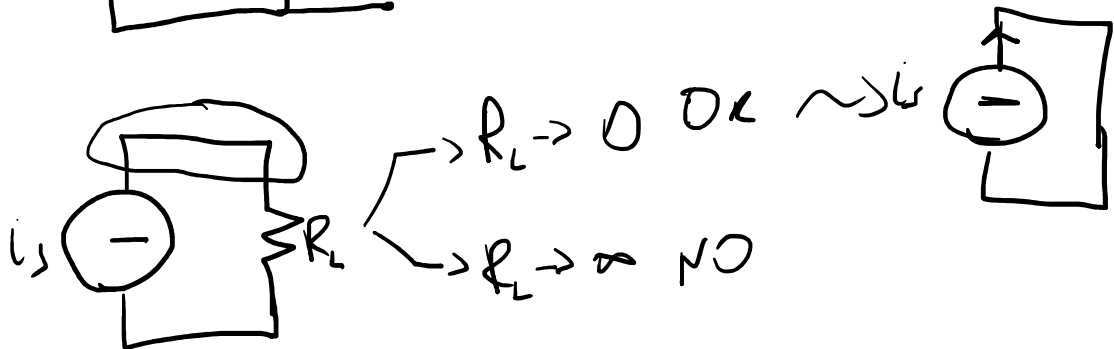
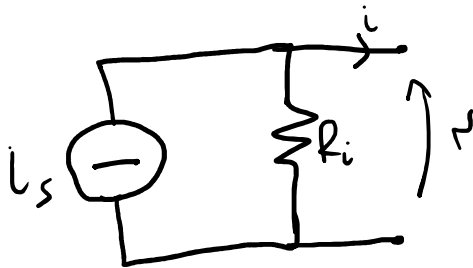
$$p(t) = v(t) \cdot i(t) = v(t) \cdot i_s$$

ATTIVO

EQ. COSTITUTIVA



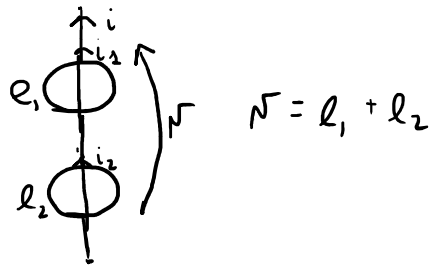
$i_s = 0 \rightarrow$  CIRCUITO APERTO



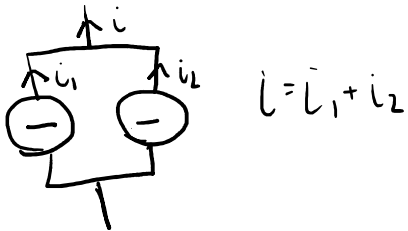
COLLEGAMENTI

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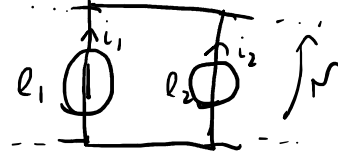
SERIE - GEN. TENSIONE



PARALLELO - GEN. CORRENTE



PARALLELO



$$U = U_1 = U_2$$

NO

NON RISPONDE

LA LKT

→ NON AMMISSIBILE

SERIE



$$i = i_1 = i_2$$

NON RISPONDE

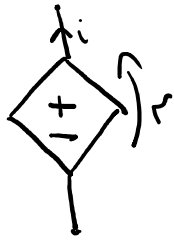
LA LKC

→ NON AMMISSIBILE

GENERATORI CONTROLLATI

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EQ. COSTIT.

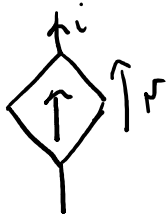


→ CONTROLLATO IN TENSIONE :

$$v = a v_{\text{con}}$$

→ CONTROLLATO IN CORRENTE :

$$v = b i_{\text{con}}$$



→ CONTROLLATO IN TENSIONE :

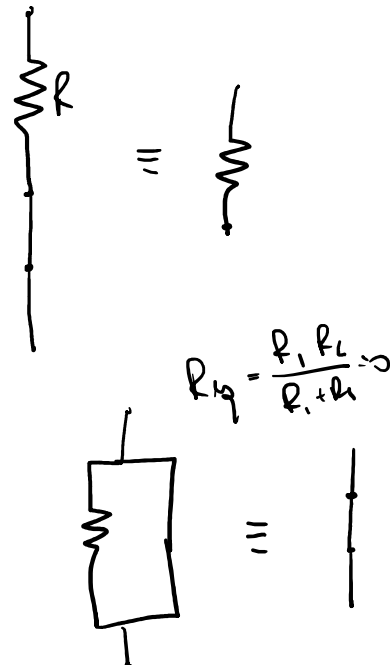
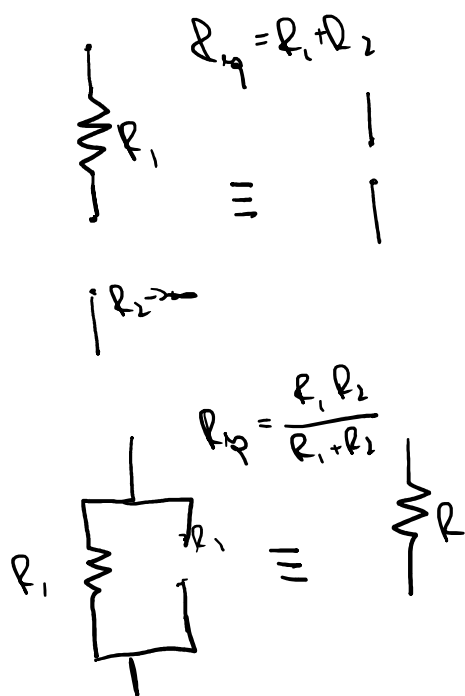
$$i = c v_{\text{con}}$$

→ CONTROLLATO IN CORRENTE :

$$i = d i_{\text{con}}$$

CONSIDERAZIONI

mercoledì 9 marzo 2022 17:47



$$G = \text{CONDUTTANZA} = \frac{1}{R} [\Omega^{-1}] [S]$$

$$\text{PARALLELO : } R_{eq} = \left(\frac{1}{R_1} + \frac{1}{R_2} \right)^{-1}$$

$$\frac{1}{R_{eq}} = R_{eq}^{-1} = \frac{1}{R_1} + \frac{1}{R_2} \rightarrow G_{eq} = G_1 + G_2$$

$$\text{SERIE : } R_{eq} = R_1 + R_2 \rightarrow \frac{1}{G_{eq}} = \frac{1}{G_1} + \frac{1}{G_2}$$

$$\rightarrow G_{eq} = \left(\frac{1}{G_1} + \frac{1}{G_2} \right)^{-1}$$